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Carbon molecular sieve membranes derived from UV-irradiated polyimides for enhanced molecular separation and physical aging resistance

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A R T I C L E I N F O

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A B S T R A C T

Herein, we present an approach for adjusting the pore structure of carbon molecular sieve (CMS) membranes by controlling UV irradiation on polyimide (PI) precursors. CMS dense membranes were fabricated by pyrolyzing UV-crosslinkable PIs synthesized from 4,4′-(hexafluoroisopropylidene)diphthalic anhydride (6FDA), 3,3′,4,4′-benzophenone tetracarboxylic dianhydride (BTDA), and 2,4-diamino mesitylene (DAM). Increasing UV irradiation time enhanced molecular sieving performance while decreasing gas permeability. At the highest UV irradiation time of 40 min, the gas selectivities for H_2/CH_4 and CO_2/CH_4 improved 3-fold and 2-fold, respectively. Meanwhile, H_2 and CO_2 permeabilities decreased by ~9 % and ~30 % relative to the pristine CMS membrane, both deviating notably from the upper bound slope trends. Gas diffusivity and pore size distribution analyses suggested that these improvements stemmed from a reduction in larger ultramicropores. Furthermore, a prolonged CO_2/CH_4 mixed-gas permeation test (170 days) showed that pre-crosslinked CMS membranes demonstrated enhanced resistance to physical aging. After the aging period, the pre-crosslinked CMS membranes showed slightly higher mixed-gas CO_2 permeability and ~2 times higher mixed-gas CO_2/CH_4 selectivity than pristine CMS. This study introduces a novel method for tuning the CMS microstructure to produce more selective and aging-resistant carbon membranes.

1. Introduction

Carbon molecular sieve (CMS) membranes exhibit superior and adjustable separation ability, ensuring their suitability for diverse membrane-based separation applications [1,2]. In addition, CMS can be produced in a practical membrane format, such as hollow fibers, providing feasible adaptability [3–6]. Numerous studies [6–8] have reported that CMS membranes typically exhibit a dual pore size distribution composed of micropores that promote high gas permeation and ultramicropores that impart molecular sieving capabilities. Specifically, Rungta et al. explained that during pyrolysis, entangled polymer chains fragment into short aromatic strands [7]. These strands tend to arrange into graphene-like plate structures. The disordered arrangement of these plates typically generates micropores that offer large gas sorption capacities and long diffusive displacement distances, contributing to fast gas permeation. In contrast, ultramicropores arise from slit-like gaps between the aromatic strands on the plates, providing molecular sieving properties [7]. Later, Hays et al. proposed the involvement of slit bypass pores in the process of physical aging [9]. These pores initially form as imperfections between adjacent graphene-like plates and, over time, develop into ultramicropore-like voids larger than typical slits, representing the higher-end range of the ultramicropore distribution [9]. Physical aging in CMS membranes is an inherently slowing process driven by the relaxation of pore volume as the membrane approaches

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thermodynamic equilibrium [10]. As the pore volume decreases over time, gas permeability typically decreases, while gas selectivity tends to increase [10]. Recent studies highlight that during physical aging, morphological changes in CMS membranes derived from polyimides primarily affect ultramicropores rather than causing micropore collapse [11]. Notably, large ultramicropores, known as slit bypass pores, are particularly important in this process. Liu et al. proposed a two-stage progression for physical aging process in CMS membranes [11]: During the initial stage, a notable reduction in gas permeation along with rise in separation properties arises from the densification of ultramicropores (i.e., orphan strands) in the continuous ("C") matrix and constriction of slit bypass pores in the Langmuir ("L") regions [11]. In the subsequent phase, once the constriction of slit bypass pores is complete, densification of the continuous phase persists, leading to gradual decreases in gas permeability while selectivity stabilizes, ultimately reaching a pseudo-equilibrium state of separation performance [11].

The gas transport characteristics in CMS membranes can be optimized through the adjustment of pyrolysis conditions [12–15], modifying polymer precursors [6,16–20], and implementing post-treatment processes [21–23]. Aromatic polyimides are widely considered promising precursors for CMS membranes, owing to their tunable structural characteristics. Furthermore, they usually exhibit high carbon yields (>50 % in N_2) [19,24]. Researchers have studied the gas permeation characteristics of CMS membranes fabricated from polyimides such as biphenyl tetracarboxylic dianhydride (BPDA), pyromellitic dianhydride (PMDA), and 4,4′-(hexafluoroisopropylidene)diphthalic anhydride (6FDA) using various diamine combinations [8,16]. Crosslinking is one effective approach for modifying the chemical structure of polyimides. Thermal and chemical crosslinking methods have been extensively employed to crosslink polyimide precursors for CMS membranes [14,17,19,20]. For instance, Fu et al. explored the gas permeation properties of CMS membranes prepared from decarboxylation-induced crosslinked 6FDA/DETDA (3:2) polyimides [14]. The pre-crosslinked CMS membranes showed improved gas permeation but slightly reduced separation properties compared to their uncrosslinked counterparts [14]. They attributed this to microvoids created during decarboxylation-induced crosslinking, which were preserved during subsequent pyrolysis [14]. Additionally, the gas transport behavior of CMS membranes fabricated from brominated 6FDA–2,4-diamino mesitylene (DAM) crosslinkable polyimide was explored by Xu et al. [18]. This approach enhanced both the gas permeability and aging resistance of the CMS membranes, primarily owing to the pre-crosslinked precursor structure, which effectively minimized pore collapse during pyrolysis [18].

As mentioned above, most studies on precursor crosslinking in CMS membranes have focused on crosslinked polymers, which generally exhibit increased free volume. This often leads to a more open pore structure, enhancing permeability but often resulting in a trade-off with selectivity. Additionally, conventional thermal treatments have been the primary method for optimizing CMS selectivity. However, alternative strategies that enable precise control over pore size and distribution are essential for further enhancing separation performance.

To address this, we introduce UV-induced precursor crosslinking as a new strategy beyond thermal treatments. Unlike conventional crosslinking, UV crosslinking promotes densification at the precursor stage, which we hypothesize may enable more precise control over the ultramicropore structure, allowing for the tuning of slit bypass pores. This, in turn, enhances molecular sieving properties and improves physical aging resistance.

This study explores the impact of UV crosslinking on gas transport properties and aging resistance, offering insights into its potential as an effective alternative for tailoring CMS membrane characteristics. UV-induced crosslinking offers several advantages, including high speed, spatial resolution, and solvent-free processing [25]. Additionally, it enables ambient temperature operation, low energy consumption, and crosslinking in solid-state films [25]. High molecular weight (M_w = 160,000 g/mol) UV-crosslinkable polyimides were synthesized from a 70/30 dianhydride molar ratio of 6FDA/3,3′,4,4′-benzophenone tetracarboxylic dianhydride (BTDA) with the diamine DAM. Upon UV irradiation, the benzophenone unit in BTDA can crosslink with a benzylic hydrogen donor (DAM) within the polyimide segment. Before pyrolysis, these polyimides underwent UV-induced crosslinking for irradiation times of 20 and 40 min. A pyrolysis condition of 625 °C with 1 h soaking time was selected based on preliminary optimization and literature trends to ensure an optimal balance between molecular sieving properties and gas permeability. The pore structures of the resulting CMS membranes were examined through gas sorption measurements and a pore size distribution analysis. Pure-gas permeability coefficients for H_2, CO_2, N_2, and CH_4 were measured at 2 atm and 35 °C. Using the measured CO_2 and CH_4 solubility coefficients, combined with the permeation data, diffusivity values were calculated to characterize the roles of sorption and diffusion in the variations in permeation and selectivity of the membranes in this study. Focused attention was also given to the long-term (e.g., 170 days) physical aging behavior of the carbon membranes with mixed-gas equimolar CO_2/CH_4 permeation behavior.

2. Experimental

2.1. Polymer synthesis and membrane casting

6FDA/BTDA–DAM polyimide was synthesized via the solution polycondensation method using a one-pot imidization process [26]. The materials information and details of synthesis procedures are provided in the supplementary information. The molecular structure of 6FDA/BTDA–DAM is presented in Fig. 1, with n = 7 and m = 3, confirmed via ^{1}H nuclear magnetic resonance (NMR) (Fig. S1). Gel permeation chromatography (GPC) determined a weight-average molecular weight of M_w = 160,000 g/mol with a dispersity of 3.7. To cast 6FDA/BTDA–DAM polymer films, 3 wt% of polyimides were solubilized in NMP at 50 °C for 3 h. The mixture was then filtered through a microfilter (2 μm PTFE) to separate residues. A 30 mL aliquot of the polyimide solution was transferred into a 14 cm diameter glass Petri dish. The following drying steps were performed in a vacuum chamber: initially, heating at 80 °C under atmospheric pressure for 48 h to evaporate the bulk NMP, followed by complete removal of the residues in a vacuum at 205 °C for 24 h.

2.2. Ultraviolet irradiation

UV irradiation was performed on the films in air using a custom-built UV crosslinker featuring a high-intensity mercury-vapor lamp optimized to irradiate predominantly at 365 nm. The UV light intensity was 265 W/m^2, measured with a radiometer (Delta OHM HD2102.2, equipped with LP 471 UVA probe). To crosslink both sides of the films, one side was irradiated for half of the overall exposure duration and then flipped over for the remaining time. The membranes were labeled according to the overall exposure duration. For instance, PI-UV20 underwent UV crosslinking for 20 min (10 min per side). Ultraviolet–visible spectroscopy (UV–Vis) was employed (Agilent Cary 5000 UV–Vis–NIR spectrophotometer) to analyze the UV absorption of the polyimide films. The power density of the UV light absorbed by the polyimide films, I_{abs}, was calculated using the Beer–Lambert law [27]:

$$I_{abs} = I_0 \times \left(1 - 10^{-A}\right) \quad (1)$$

where I_0 is the UV light intensity at a specific wavelength in W/m^2, and A is the absorbance of the sample film. The irradiation dose (D) can be estimated using Eq. (2):

$$D = I_{abs} \times t / \rho_A \quad (2)$$

where t presents the UV exposure duration in seconds, and ρ_A is the area

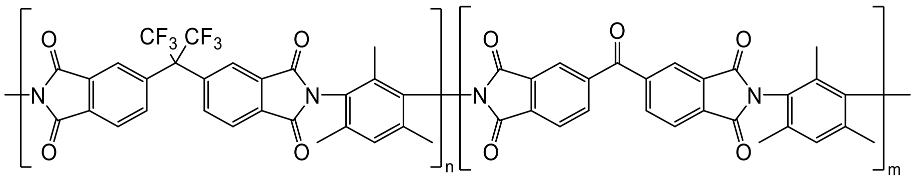

Fig. 1. Chemical structure of 6FDA/BTDA–DAM polyimide (n = 7, m = 3).

density of samples in kg/m^2. The irradiation doses for PI-UV20 and PI-UV40 are 4078 and 8155 kGy, respectively.

2.3. Preparation of CMS membranes

Uncrosslinked and UV-treated 6FDA/BTDA–DAM films were pyrolyzed in a custom-built tubular furnace with a controlled temperature profile to obtain CMS membranes. The furnace was flushed with ultrahigh-purity helium flowing at 50 mL/min, and an oxygen level below 10 ppm was maintained throughout the process, as confirmed by the RapidOx 2100 (Cambridge Sensotec, UK). The heating rate, final pyrolysis temperature, and soaking time were 3 °C/min, 625 °C, and 1 h, respectively. The CMS membranes were removed after the temperature naturally cooled to ambient conditions. The overall pyrolysis protocol is presented in Fig. S2. The thickness of the CMS films was ~50 μm.

2.4. Characterization

^{1}H NMR spectroscopy was used to verify the molecular structure of 6FDA/BTDA–DAM polyimide. The NMR spectra were obtained using a Bruker AVANCE 500 MHz ^{1}H NMR spectrometer (Bruker, Billerica, USA) in $CDCl_3$ at room temperature.

The weight-average molecular weight (M_w) with a dispersity of 6FDA/BTDA–DAM was obtained using GPC. The system included an autosampler (NS-6000, Futecs, Korea), a refractive index detector (VE 3580, Viscotek, UK), and two columns of PLgel Mixed-C and Mixed-D (Agilent, USA) connected in series, chloroform as the mobile phase. Calibration curves were obtained using polystyrene standards (Shodex, Korea).

Thermogravimetric analysis (TGA) was performed using a Thermal Analysis System TGA 2 (Mettler Toledo, USA) applying a heating ramp of 5 °C/min from 30 to 800 °C. A 50 mL/min N_2 purge flow rate was supplied to the TGA balance and sample compartment, respectively.

Differential scanning calorimetry (DSC) was conducted using DSC 214 (Netzsch, Germany), heating polymer films from 100 to 450 °C at 10 °C/min in a N_2 atmosphere. The glass transition temperature (T_g) of the membrane films was assessed through the subsequent heating cycle.

Gel fraction was used to estimate the extent of crosslinking in UV-irradiated polymer films. Samples were immersed in NMP for 24 h at ambient temperature. The non-dissolvable content was subsequently vacuum-dried at 205 °C for 24 h before measurement.

The density of all polymer films was measured using an AD-1653 density determination kit (A&D Company, Japan), with *n*-heptane (Sigma-Aldrich, ReagentPlus grade, density: 0.684 g/cm^3 at 20 °C) as the reference liquid.

The fractional free volume (FFV) of polymer films was estimated using Eq. (3) [28]:

$$FFV = \frac{V - V_0}{V} \tag{3}$$

where V represents the reciprocal of the polymer density, and V_0 represents the occupied volume of the polymer segments [28], defined as $V_0 = 1.3 \sum V_W$, where V_W is the theoretical volume (van der Waals volume) determined from the group contribution method [29].

Raman spectroscopy was used to investigate the structure of CMS films derived from PI precursors with and without UV irradiation. The measurements were performed on a SENTERRA spectrometer (Bruker, Billerica, USA). The laser wavelength and power were 532.04 nm and 1.44 mW, respectively.

X-ray photoelectron spectroscopy (XPS, Kratos AXIS Nova, Wharfside, UK) was performed to analyze the elemental composition of CMS films. Monochromatic Al Kα source was used, and the accelerating voltage was set to 15 keV.

Low-pressure CO_2 adsorption of 6FDA/BTDA–DAM CMS was performed at 298 K using a Belsorp Max II (MicrotracBEL, Japan), with the proportional pressure (p/p_0) spanning from 1.0×10^{-6} to 0.99. The CMS films were finely ground using an agate mortar before the CO_2 adsorption tests. Grand Canonical Monte Carlo simulations were adopted to analyze the pore size distribution, as implemented in the equipment, assuming a slit-pore geometry for carbon with ultramicropores.

2.5. Pure- and mixed-gas permeation measurements

Before gas permeation tests, all polymer and CMS films were masked with a ring-shaped copper plate using epoxy (Devcon #20845, USA). The effective membrane film surface was 0.4–0.7 cm^2. The constant volume-variable pressure protocol [30] was applied to measure pure-gas permeability for polymer films at 35 °C with an upstream pressure of 2 atm. Downstream pressure was measured with a 2 Torr capacitance transducer (Baratron, 626B, MKS, Andover, MA, USA). In this study, pure H_2 and CO_2 permeation through CMS films was too high to achieve sufficient steady-state pressure rate in the downstream using our constant volume-variable pressure system. To ensure measurement consistency, we applied the constant pressure-variable volume protocol [30] for all CMS membranes. In addition, CMS membranes with low pure gas permeation properties and mixed-gas permeation properties were measured using helium sweep gas with at a flow rate of 1 sccm to improve the data accuracy. Pure-gas measurements were carried out at an upstream pressure of 2 atm at 35 °C. Mixed-gas permeation was tested at an upstream pressure of 4 atm using an equimolar CO_2/CH_4 mixture at 35 °C. In the mixed-gas experiments, the stage cut was controlled below 0.01 to minimize concentration polarization at the membrane surface. For the downstream gases, the flow rate and concentration were determined using a flow gauge and a gas chromatograph (DS6200, Donam Instruments, Korea) incorporating a thermal conductivity detector, respectively.

2.6. Gas sorption measurements

A pressure-decay protocol was used to characterize CO_2 and CH_4 sorption in CMS films [30]. This system consists of a dual-volume and dual-transducer. [30]. The system comprises two cells (e.g., charge and sample) with known volumes. The Burnett expansion protocol was applied to measure the cell volumes [31]. Using two pressure transducers (PMP5074 series, GE, USA), the measurements were performed over a pressure range of 2–14 atm in an increasing manner to obtain sorption isotherms. OriginLab® software was employed to fit the measured data to the dual-mode sorption model through an optimal nonlinear fitting technique. The solubility coefficients (S) were determined from the experimental sorption results using the following equation [32]:

$$S = \frac{C}{p} \quad (4)$$

where C is the concentration of sorbed gas in membrane samples, and p is pressure (atm).

3. Results and discussion

3.1. Characterization of polymer precursors

Upon exposure to UV light (250–370 nm), the most widely accepted crosslinking mechanism involves hydrogen abstraction [33,34]. Previous studies have reported UV-induced crosslinking of benzophenone-containing polyimides used in membrane-based gas separation [33–35]. In the case of 6FDA/BTDA–DAM, as illustrated in Fig. 2 [34], the crosslinking reaction begins when the carbonyl group of the benzophenone moiety in BTDA is activated by UV light, generating a radical. This radical then abstracts a reactive hydrogen atom from a methyl component attached in DAM, forming a covalent bond [33–35]. This process may lead to densification, reducing the polymer free volume (reaction A). Additional crosslinking can occur between polymer segments when the radicals on the benzylic positions react (reaction B). Furthermore, UV exposure under air often induces photo-oxidation on the surface, potentially reducing the free volume due to additional densification [35,36].

Table 1 presents the physical and thermal characteristics of uncrosslinked and UV-treated 6FDA/BTDA–DAM polyimides in this study. The label "PI" refers to the polyimide, and the number following "UV" indicates the irradiation time (min). The membrane films were exposed to UV irradiation on both surfaces. For example, PI-UV40 denotes a PI film irradiated for 20 min on each side. As shown in Table 1, the extent of crosslinking in the irradiated PIs is estimated from their gel fraction. The uncrosslinked PI sample dissolves readily in the casting solvent, while the gel fractions for the irradiated samples, PI-UV20 and PI-UV40, are 93 % and 97 %, respectively. UV light intensity typically decreases with depth within the films, leading to a non-uniform distribution of crosslinking [34,35,37]. However, the irradiation dose in this study was sufficient to markedly reduce the polymer free volume, in agreement with previous studies [33–35]. As the irradiation time increases, the fractional free volume (FFV) decreases from 17.3 % (PI) to 12.8 % (PI-UV40), indicating densification via crosslinking and photo-oxidation. The T_g of UV-irradiated PIs increases slightly with irradiation time, suggesting the development of additional bonds between polymer segments, resulting in a rigid network and restricted chain mobility [35]. TGA scans of uncrosslinked and UV-treated PIs are shown in Fig. S3. The TGA curves for all three samples are nearly identical, indicating that UV-induced crosslinking has minimal effect on the thermal properties of the polymers, consistent with previous studies [34]. No thermal degradation is observed below 300 °C, and the final weight loss, corresponding to the carbon yield, is ~55 %.

Fig. 3 presents the relative permeability coefficients of uncrosslinked and UV-treated PIs with respect to UV irradiation time. The relative permeability is expressed as the proportion of the permeability of the irradiated PI to that of the uncrosslinked PI. Consistent with the decrease in FFV values, the gas permeability coefficients for all gases decrease with increasing UV irradiation time. For all polymers, permeability follows a decreasing order as follows:

H_2 (2.89 Å) > CO_2 (3.30 Å) > N_2 (3.64 Å) > CH_4 (3.8 Å)

The kinetic diameters are represented by the values in parentheses [38]. As shown in Fig. 3, the relative permeability of the irradiated samples decreases with increasing UV irradiation time, with a highly pronounced effect on larger gas molecules. This trend is consistent with the UV crosslinking and photo-oxidation processes, which reduce the average free volume due to polymer densification [33–35,38]. Consequently, gas selectivity increases substantially with extended UV irradiation time. For example, for the PI irradiated for 40 min, H_2/CH_4 and CO_2/CH_4 separations increase by factors of ~35 and ~4, respectively. Therefore, the physical properties and gas transport characterization confirm that the UV-irradiated PIs, used as precursors for CMS membranes, are highly crosslinked and structurally distinct from the uncrosslinked PI.

3.2. Characterization of carbon materials

3.2.1. Raman spectroscopic analysis

Fig. 4 presents the Raman spectra of CMS films derived from uncrosslinked PI (CMS PI) and UV-treated PIs (CMS PI-UV20 and CMS PI-UV40). Raman spectroscopy provides valuable insights into the structure of amorphous carbons, including carbon molecular sieves [39,40]. While the two characteristic peaks, the "G" band and "D" band, are widely used to understand the structure of CMS, the Raman spectra in this study were additionally decomposed to resolve peaks using CasaXPS software [41]. Detailed parameters of the decomposed peaks are summarized in Table 2. The G peak at ~1585 cm^{-1} arises from vibrations with E_{2g}-symmetry in a perfect carbon lattice structure, while the D1 peak at ~1350 cm^{-1} originates from vibrations with A_{1g}-symmetry in an imperfect carbon network, indicating planar defects or boundaries of graphene layers [42]. The D2 peak at ~1610 cm^{-1} is speculated to represent vibrations with E_{2g}-symmetry of detached graphene layers excluded from the crystalline carbon structure [42]. The D3 peak at

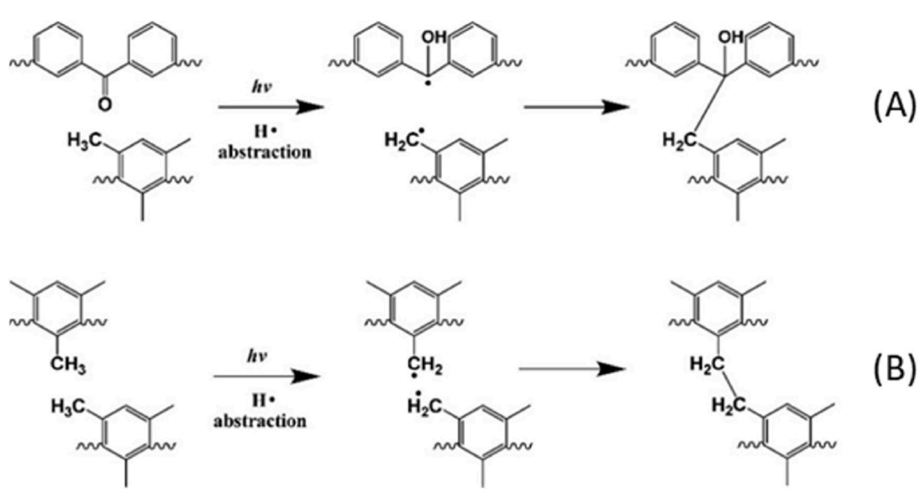

Fig. 2. Proposed pathway for UV-induced crosslinking [34].

Table 1
Physical and thermal characteristics of uncrosslinked and UV-treated 6FDA/BTDA–DAM polyimides.

Sample	Irradiation time (min)	Irradiation dose (kGy)	Gel fraction	T_g (°C)	Density (g/cm^3)	FFV
PI	–	–	–	376	1.313 ± 0.006	0.173
PI-UV20	20	4078	93 %	378	1.354 ± 0.009	0.147
PI-UV40	40	8155	97 %	387	1.384 ± 0.010	0.128

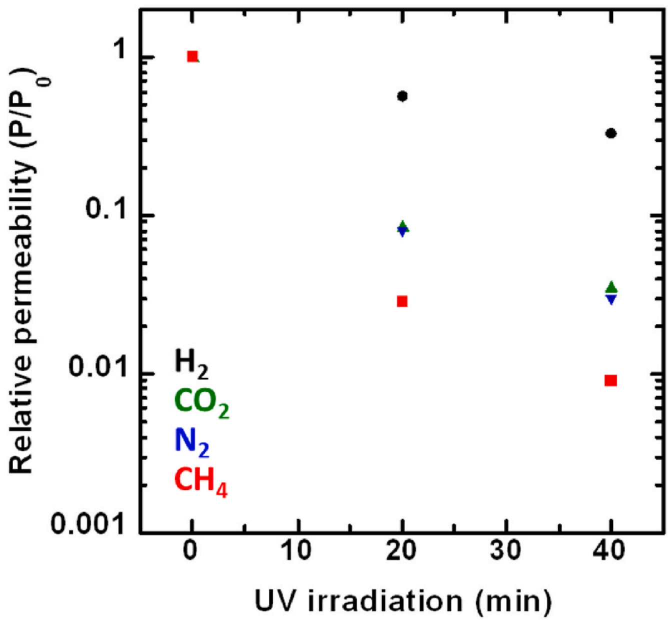

Fig. 3. Relative permeability at 2 atm and 35 °C for 6FDA/BTDA–DAM polyimides with respect to irradiation time (min). Permeability coefficients are also listed in Table 4.

~1500 cm^{-1} and the D4 peak at ~1235 cm^{-1} represent amorphous carbon components: the D3 peak is associated with random mixtures of sp^2 and sp^3 hybridized carbons, while the D4 peak corresponds to sp^2-sp^3 C − C & C = C stretching vibrations located at the edges of graphitic lattice [42–44]. Furthermore, the method proposed by Ferrari and Robertson [42,45] was used to determine the sp^3/sp^2 carbon ratio of CMS films.

As shown in Table 2, a trend of decreasing A_{D1}/A_{all} and increasing A_{D3}/A_{all} is observed with increasing UV irradiation time applied to the PI precursor (note: A_{band} = area of deconvoluted band). The reduction in the A_{D1}/A_{all} ratio is likely owing to the densification of the PI precursor structure induced by UV crosslinking prior to carbonization. As detailed in Table 1, UV irradiation increases the density of the PI precursor while reducing its FFV. We presume that this densification promotes better coalescence of polymer chains during carbonization, leading to a reduction in defects associated with in-plane graphitic boundaries. On the other hand, the increase in the A_{D3}/A_{all} ratio suggests a higher proportion of amorphous carbon with random sp^2 and sp^3 hybridized bonds. This increase reflects the restricted chain mobility caused by UV-induced crosslinking (cf. Section 3.1), which hinders the proper alignment of polymer chains into extended graphitic crystalline structures. This interpretation is reinforced by the observed increase in the sp^3/sp^2 hybridized carbon ratio with UV irradiation time. The spatially distributed sp^3-hybridized carbons formed during crosslinking presumably disrupt the development of ordered graphitic domains, compared to those in pristine CMS PI membranes [21].

In addition to the Raman spectroscopy, X-ray photoelectron spectroscopy (XPS) was carried out to further elucidate the carbon bonding environment in CMS membranes (cf. Figs. S4 and S5, Tables S1 and S2). The C 1*s* narrow scan spectra exhibit a substantial increase in sp^3 carbon contributions for CMS films derived from UV-irradiated PI precursors, consistent with the trends in the Raman-derived sp^3/sp^2 ratios. This further supports the idea that UV crosslinking induces a more disordered carbon matrix with higher fractions of sp^3 carbon, thereby reinforcing the conclusion that UV pre-treatment hinders formation of ordered graphitic domain in the final CMS membranes.

3.2.2. *Pore size distribution analysis*

Fig. 5 shows the pore size distribution of the CMS membranes, which exhibit a typical bimodal structure (e.g., micropores and ultramicropores) [7,8,20]. Note that the pore size distribution was measured via low-pressure CO_2 adsorption. Given CO_2's kinetic diameter (e.g., ~3.3 Å), this method probes adsorbed gas rather than directly

Table 2
Deconvoluted Raman band parameters of CMS membranes.

Samples	Band area ratio[a]					sp^3/sp^2 carbon ratio[b]
	A_{D1}/A_{all}	A_{D2}/A_{all}	A_{D3}/A_{all}	A_{D4}/A_{all}	A_G/A_{all}	
CMS PI	0.265	0.043	0.321	0.208	0.163	0.338
CMS PI-UV20	0.245	0.043	0.341	0.215	0.156	0.362
CMS PI-UV40	0.215	0.049	0.360	0.225	0.151	0.388

[a] $A_{all} = A_{D1} + A_{D2} + A_{D3} + A_{D4} + A_G$

[b] sp^3/sp^2 carbon ratio $= \dfrac{0.2A_{D3} + 0.85A_{D4}}{A_{D1} + A_{D2} + A_G + 0.2A_{D3} + 0.85A_{Dd4}}$ [45].

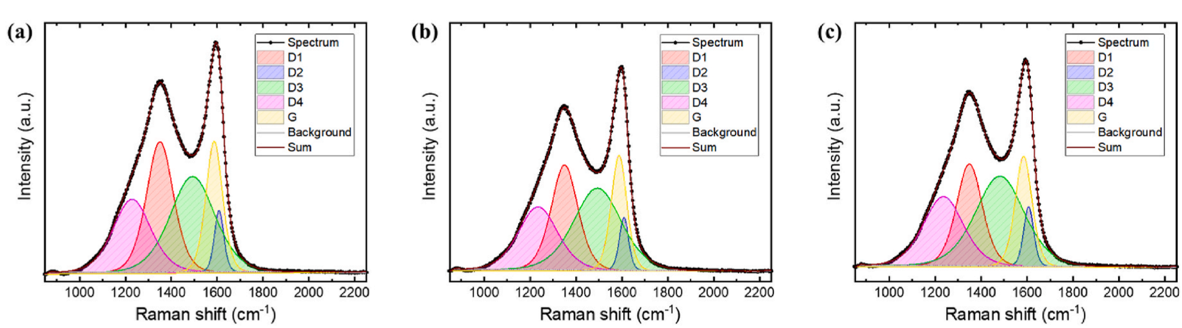

Fig. 4. Deconvoluted Raman spectra of (a) CMS PI, (b) CMS PI-UV20, (c) CMS PI-UV40 membranes, pyrolyzed at 625 °C for 1 h. The spectra are decomposed to resolve peaks (D1, D2, D3, D4, and G).

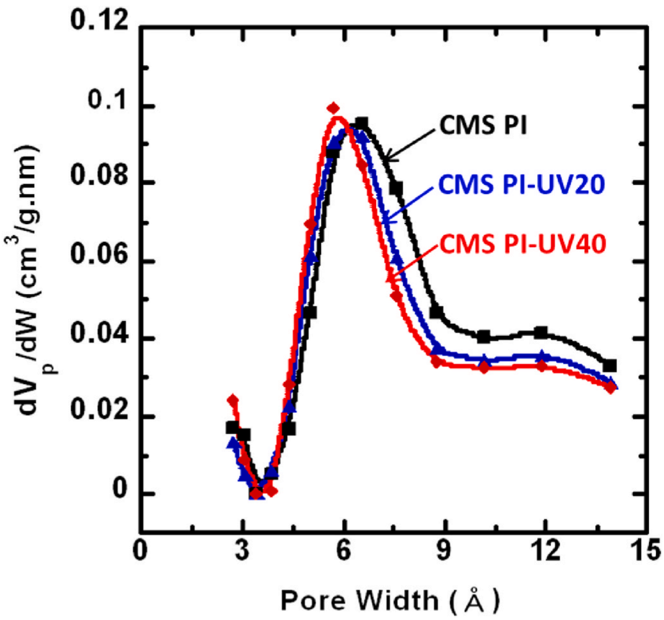

Fig. 5. Pore size distribution of CMS membranes derived from uncrosslinked and UV-treated PI precursors, pyrolyzed at 625 °C for 1 h.

measuring pores, which generally results in an overestimation [4,46,47]. Thus, the data are interpreted qualitatively for relative comparison of CMS materials. Notably, CMS membranes derived from UV-irradiated samples (e.g., CMS PI-UV20 and CMS PI-UV40) exhibit a different pattern in the ultramicropore distribution compared to pristine CMS (e.g., CMS PI), while the micropore distribution remains relatively unchanged. For example, the ultramicropore distribution in CMS PI-UV20 and CMS PI-UV40 narrows, notably on the right side of the range. Additionally, the upper end of the ultramicropore range is associated with slit bypass pores, as described by Hays et al., which result from edge imperfections between neighboring ultramicroporous plates [9]. This narrowing of the ultramicropore distribution suggests that UV-induced crosslinking of the PI precursors effectively tunes the formation of ultramicropores, particularly reducing larger ultramicropores, including slit bypass pores, thereby influencing the penetration of relatively larger gas molecules. Furthermore, this narrowing of larger ultramicropores is consistent with the decreasing trend in A_{D1}/A_{all} observed in Table 2. Further pore structure analysis, along with gas sorption results, is presented in Section 3.2.3.

3.2.3. Pure-gas sorption properties

Fig. 6 shows pure-gas sorption isotherms for CO_2 (Fig. 6a) and CH_4 (Fig. 6b) in CMS membranes derived from uncrosslinked and UV-treated samples. The CO_2 and CH_4 sorption isotherms show similar trends across all CMS membranes. The measured values were fitted to the dual-mode sorption model [2,28]. The obtained parameters are summarized in Table 3. As reported in previous studies [2,6,11], CMS membranes derived from PI precursors exhibit a dual-mode sorption behavior as described in Eq. (5):

$$C = k_C p + C'_L \frac{bp}{1 + bp} \tag{5}$$

where p is pressure, k_C is the sorption constant of the continuous mode, C'_L is the Langmuir capacity constant, and b is the affinity constant. This consists of a minor disordered continuous phase ("C"), which follows Henry's law of sorption, and a majority microporous domain ("L"), governed by Langmuir sorption. The "C" phase is attributed to carbon strands that fail to organize into graphene-like platelets due to limited space and steric hindrance [2,6,11]. These carbon strands are assumed to be randomly packed around the plate-like and cellular structures, resulting in the disordered continuous ("C") phase [2,6,11].

As most gas sorption occurs in the micropores, the sorption isotherms can also effectively probe the micropore structure of CMS membranes [2,14,16,48]. As shown in Table 3, for all CMS membranes, the values of C'_L and b are similar for each gas, indicating that UV crosslinking of the PI precursors has minimal impact on the micropore structure and distribution. Additionally, the relatively unchanged solubility coefficients for all CMS membranes reveal that any variations in gas permeation

Table 3
Dual-mode sorption parameters for CO_2 and CH_4 in CMS membranes derived from uncrosslinked and UV-treated polyimide precursors.

Gas	Samples	Dual-mode sorption parameters		
		C'_L (cm^3(STP)/(cm^3 CMS))	b (atm^{-1})	k_C (cm^3(STP)/(cm^3 CMS·atm))
CO_2	CMS PI	119	0.60	3.7
	CMS PI-UV20	122	0.64	3.7
	CMS PI-UV40	121	0.63	3.4
CH_4	CMS PI	73	0.34	2.3
	CMS PI-UV20	71	0.33	2.1
	CMS PI-UV40	66	0.31	2.0

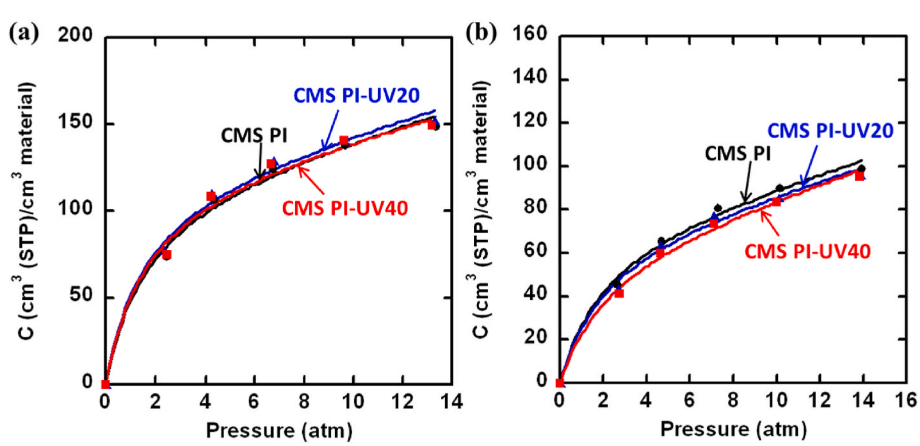

Fig. 6. Sorption isotherms at 35 °C of (a) CO_2 and (b) CH_4 in CMS membranes derived from uncrosslinked and UV-treated polyimide precursors, pyrolyzed at 625 °C for 1 h.

Table 4
Pure-gas permeability and selectivity at 2 atm and 35 °C for polymer and CMS membranes investigated in this study. The CMS films had a thickness of ~50 μm.

Sample	Pure-gas permeability (Barrer)				Pure-gas selectivity	
	H_2	CO_2	N_2	CH_4	H_2/CH_4	CO_2/CH_4
PI	430 ± 28	470 ± 33	23 ± 1.5	21 ± 1.4	21 ± 2	22 ± 2
PI-UV20	240 ± 12	40 ± 2	1.8 ± 0.08	0.6 ± 0.03	400 ± 30	66 ± 5
PI-UV40	140 ± 7	16 ± 1	0.67 ± 0.03	0.19 ± 0.01	730 ± 50	87 ± 8
CMS PI	16000 ± 1300	10000 ± 890	530 ± 50	380 ± 35	42 ± 5	26 ± 4
CMS PI-UV20	15100 ± 1100	8200 ± 590	340 ± 27	160 ± 10	96 ± 10	52 ± 6
CMS PI-UV40	14900 ± 960	6800 ± 480	230 ± 16	110 ± 7	133 ± 12	61 ± 7

Note: uncertainties were estimated using the propagation of errors method [49].

results stem primarily from changes in gas diffusion.

3.2.4. Pure-gas permeation properties

Fig. 7 presents the pure-gas permeability with respect to kinetic diameter for CMS dense membranes (~50 μm) derived from both uncrosslinked and UV-treated PI precursors. The gas permeability coefficients and selectivity values are also summarized in Table 4. As the UV irradiation time increases for the PI precursors, the gas permeability of the resulting CMS membranes decreases. As discussed previously, the micropores in CMS derived from UV-treated PIs remain largely

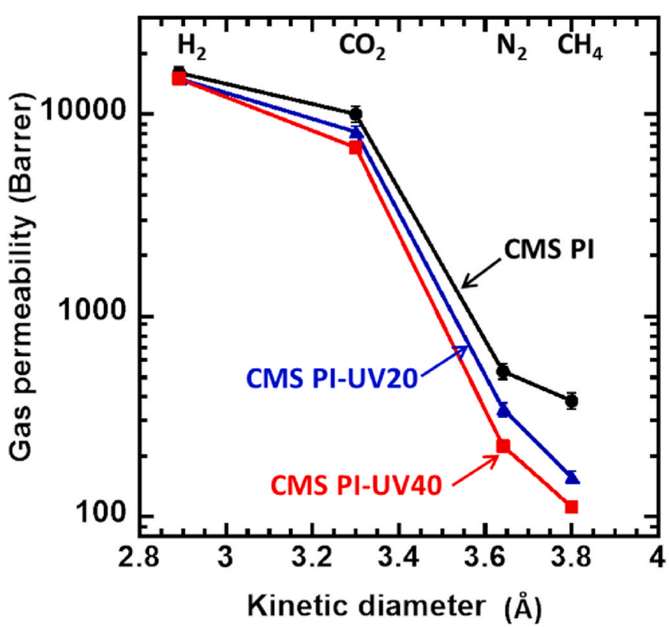

Fig. 7. Pure-gas permeability at 2 atm and 35 °C for CMS membranes derived from uncrosslinked and UV-treated PI precursors, pyrolyzed at 625 °C for 1 h. The CMS films had a thickness of ~50 μm. Lines are provided for visual reference.

unchanged. The decrease in gas permeability is primarily ascribed to the reduction in ultramicropores. Furthermore, the relative decrease in gas permeability is highly pronounced for larger gas molecules (e.g., ~9 % reduction for H_2 and ~70 % for CH_4 in CMS PI-UV40), suggesting the development of more size-discriminating ultramicropores. Consequently, as Table 4 shows, the gas selectivity of CMS membranes derived from UV-treated PIs increases systematically with extended UV exposures. For example, in the CMS PI-UV40 sample, H_2/CH_4 and CO_2/CH_4 selectivities increase by factors of ~3 and ~2, respectively, compared to the CMS PI sample. The observed trend is likely owing to chain packing in the PI precursors. Herein, UV exposure leads to substantial densification of PI precursors through intrachain and interchain crosslinking, driven by hydrogen abstraction, as confirmed by FFV and gas transport results (Table 1 and Fig. 3). We hypothesize that the pre-crosslinked PI structure is preserved during pyrolysis, restricting the formation of considerably larger ultramicropores, as indicated by the narrowing of the ultramicropore distribution (Fig. 5). In contrast, CMS membranes derived from decarboxylation-induced crosslinked PIs have shown an opposite trend [14], where the crosslinking results in a structure with larger pores in the CMS membranes, enhancing permeability while slightly reducing selectivity.

To further validate the enhanced size-sieving performance of CMS membranes derived from irradiated PIs, CO_2 and CH_4 diffusivity coefficients (D) were calculated using the solution-diffusion model [6,32,50] on the basis of their measured permeability (P) and solubility (S) values. Table 5 presents P, D, and S values for CO_2 and CH_4 at 2 atm and 35 °C. As noted earlier, CO_2 and CH_4 solubility remain largely unchanged across all CMS membranes. Their diffusion coefficients decrease with increasing irradiation time for the membrane precursors, indicating that the observed decreases in permeability are primarily diffusion-related. Additionally, as shown in Table 6, CO_2/CH_4 diffusivity selectivity increases systematically with the UV irradiation of the precursors, essentially mirroring the relative increase in pure-gas selectivity data. These results align with findings for CMS membranes from irradiated PIs, where pre-crosslinking in this study tightens the ultramicropores.

Fig. 8 presents the gas transport properties of H_2/CH_4 (Fig. 8a) and CO_2/CH_4 (Fig. 8b) in CMS membranes derived from uncrosslinked and UV-treated PIs (red symbols), plotted on the upper bound plot [51–54]. For comparison, gas transport data for CMS PI membrane pyrolyzed at an elevated temperature of 700 °C for 1 h is also included (blue symbol). Additionally, the gas transport properties of various carbon membranes (grey symbols) are also compared for reference. The gas transport results

Table 5
Gas permeability, solubility, and diffusivity at 2 atm and 35 °C for CMS membranes derived from uncrosslinked and UV-treated polyimide precursors, pyrolyzed at 625 °C for 1 h.

Sample	Permeability (Barrer)		Solubility ($cm^3(STP)/(cm^3$ CMS·atm))		Diffusivity x 10^9 (cm^2/s)	
	CO_2	CH_4	CO_2	CH_4	CO_2	CH_4
CMS PI	10000 ± 890	380 ± 35	30 ± 1.3	17.5 ± 0.8	25100 ± 2500	1650 ± 170
CMS PI-UV20	8200 ± 590	160 ± 10	32 ± 1.2	16.6 ± 0.6	19200 ± 1600	720 ± 50
CMS PI-UV40	6800 ± 480	110 ± 7	31 ± 1.3	15.1 ± 0.7	16600 ± 1400	570 ± 40

Note: uncertainties were estimated using the propagation of errors method [49].

Table 6

CO_2/CH_4 permeability, diffusivity, and solubility selectivity at 2 atm and 35 °C for CMS membranes derived from uncrosslinked and UV-treated polyimide precursors, pyrolyzed at 625 °C for 1 h. The CMS films had a thickness of ~50 μm.

Sample	P_{CO_2}/P_{CH_4}	S_{CO_2}/S_{CH_4}	D_{CO_2}/D_{CH_4}
CMS PI	26 ± 4	1.7 ± 0.11	15 ± 1.3
CMS PI-UV20	52 ± 6	1.9 ± 0.1	26 ± 2.3
CMS PI-UV40	61 ± 7	2.0 ± 0.12	31 ± 2.9

Note: uncertainties were estimated using the propagation of errors method [49].

and detailed information about these membranes, as well as other reported carbon membranes, are available in the supplementary information (Table S3).

Typically, for many studied CMS membranes fabricated from PI polymers, raising the final pyrolysis temperature causes a reduction in gas permeability but enhanced gas selectivity, often following the upper bound trade-off slope [8,14,55,56]. This behavior is primarily due to the more tightly packed structure of both micropores and ultramicropores with increasing pyrolysis temperature [8,14]. When comparing the gas transport performance of CMS membranes prepared at 625 °C and 700 °C in this study, similar trends are observed. However, for CMS membranes derived from UV-irradiated PIs, increasing UV irradiation time remarkably enhances H_2/CH_4 and CO_2/CH_4 selectivity with only a slight impact on H_2 and CO_2 permeability. This results in a noticeable deviation from the upper bound trend. The deviation is primarily attributed to the minimal changes in micropores, along with the compaction of larger ultramicropores, which restrict diffusion of larger penetrants (e.g., N_2 and CH_4) while facilitating the diffusion of small ones. Similar trends for H_2/N_2 and CO_2/N_2 are also observed, as shown in the supplementary information (Fig. S6). These observations highlight the enhanced permeability/selectivity combination of CMS membranes from UV-irradiated PIs, demonstrating UV crosslinking as an effective method to improve CMS performance, distinct from conventional thermal treatments.

3.2.5. Physical aging behaviors

In CMS, physical aging is a self-limiting phenomenon that results from the relaxation of pore volumes as the materials approach a thermodynamically stable state [10]. As pore volume decreases during physical aging, gas permeability typically drops while selectivity increases [10]. Recent studies suggest that, during the process of physical aging, structural changes in CMS primarily affect the ultramicropores, instead of reducing the micropores [11]. In particular, slit bypass pores, which arise from edge imperfections between adjacent plates, are readily tightened during physical aging [9,11].

Fig. 9 presents the influence of physical aging on mixed-gas absolute (Fig. 9a) and relative (Fig. 9b) CO_2 permeability, and mixed-gas CO_2/CH_4 separation properties, plotted on the upper bounds (Fig. 9c) for CMS PI and CMS PI-UV40 dense membranes (~50 μm). The gas transport

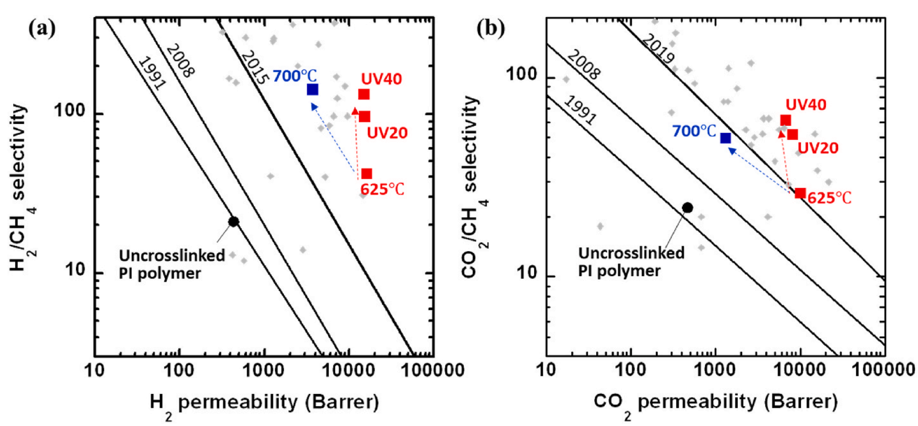

Fig. 8. Permeability/selectivity properties of (a) H_2/CH_4 and (b) CO_2/CH_4 for CMS membranes derived from uncrosslinked and UV-treated PI precursors (red symbols). Transport data for CMS PI prepared at 700 °C for 1 h (blue symbol) and uncrosslinked PI polymer (black symbol) are included for comparison. Blue and red lines are provided for visual reference. The thickness of the CMS films was ~50 μm. The gas transport results and detailed information about various carbon membranes (grey symbols) for reference are available in the supplementary information (Table S3). Error bars are smaller than the markers for data points. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

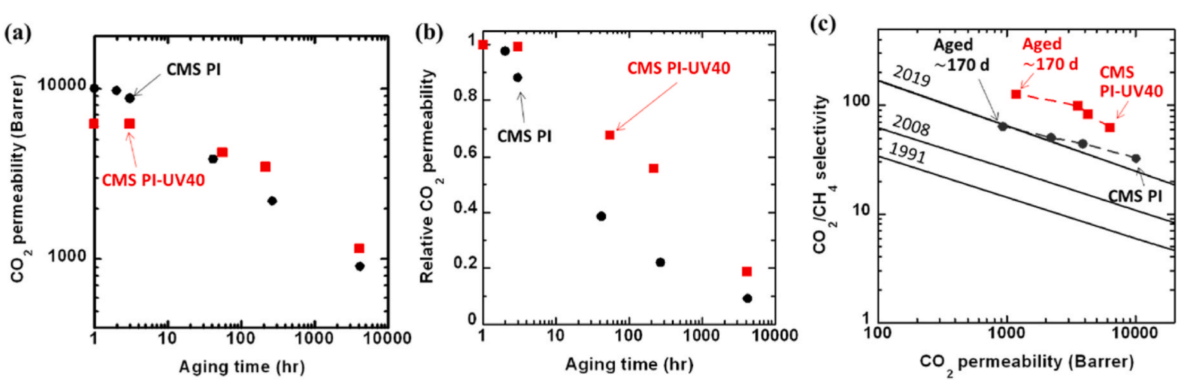

Fig. 9. Influence of physical aging on (a) absolute and (b) relative mixed-gas CO_2 (50 %) permeability and (c) mixed-gas CO_2/CH_4 (50/50) separation performance with respect to the upper bounds for CMS PI and CMS PI-UV40 samples, pyrolyzed at 625 °C for 1 h. The CMS film thickness was ~50 μm. Lines are provided for visual reference. Error bars are smaller than the markers for data points.

data shown in Fig. 9 are also listed in Table S4. Measurements were conducted up to ~4100 h (~170 days). The membranes were kept in a vacuum between measurements to minimize exposure to oxygen and water vapor. Note that vacuum treatment also accelerates physical aging in CMS membranes [11]. For the CMS PI sample, as shown in Fig. 9b, the initial measurement, taken 40–50 h after vacuum exposure, shows a profound decrease in CO_2 permeability, followed by a slow decline over the remaining aging period. This rapid aging is likely due to simultaneous narrowing of slit bypass pores in the Langmuir regions and ultramicropore densification in the continuous phase, as suggested by Liu et al. [11]. CO_2 permeability in the CMS PI-UV40 sample also decreases noticeably during the experiment but to a lesser extent than in CMS PI. After ~4100 h of aging, CO_2 permeability in CMS PI-UV40 is ~30 % higher than in CMS PI, with values of 1200 and 925 Barrer, respectively. Although physical aging is not complete for either sample within the tested period, these results suggest that pre-crosslinking of the PI precursor delays physical aging. The relatively small reduction in CO_2 permeability for CMS PI-UV40 is likely attributed to the pre-tightened ultramicropores, particularly slit bypass pores, which limit further contraction during the early stages of physical aging. As presented in Fig. 9c, physical aging increases CO_2/CH_4 mixed-gas selectivity for both samples. After ~4100 h (~170 days) of aging, the CMS PI-UV40 sample shows slightly higher CO_2 permeability and approximately twice the CO_2/CH_4 separation performance relative to the CMS PI sample, indicating an enhanced combination of permeability and selectivity.

4. Conclusions

This study demonstrated that UV-induced crosslinking of the 6FDA/BTDA–DAM precursor considerably enhanced the gas transport performance in the derived CMS membranes. UV-induced crosslinking formed a denser polymer structure, influencing the development of ultramicropores. This process likely tuned the ultramicropore network, particularly reducing larger ultramicropores, including slit bypass pores. Consequently, CMS membranes derived from UV-irradiated PIs exhibited improved molecular sieving properties without notably compromising the permeability of small gases (e.g., H_2 and CO_2) compared to pristine CMS membranes, providing an alternative to conventional thermal treatments. Additionally, pre-crosslinking appears to delay physical aging, allowing UV-treated CMS membranes to maintain higher CO_2 permeability and CO_2/CH_4 selectivity than pristine CMS membranes after aging. These findings, consistent with recent reports, highlight the importance of tuning larger ultramicropores to enhance molecular sieving and resistance to physical aging. Furthermore, UV irradiation of the polyimide precursor offers a promising approach for optimizing CMS membrane performance and developing high-performance gas separation membranes with improved stability.

CRediT authorship contribution statement

Daehun Kim: Investigation, Writing – original draft, Writing – review & editing. **Mi-Hee Ryu:** Investigation, Writing – original draft, Writing – review & editing. **Ahrumi Park:** Conceptualization, Investigation. **Joo-Eon Kim:** Investigation. **Seong-Joong Kim:** Writing – review & editing. **YongSung Kwon:** Investigation, Writing – review & editing. **Jungkyu Choi:** Funding acquisition, Supervision, Writing – review & editing. **Jaesung Park:** Funding acquisition, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.memsci.2025.124080.

Data availability

No data was used for the research described in the article.

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